

Shortforms

F.C - Front Controller
D.S - DispatcherServlet
H.M - HandlerMapping bean
V.R - ViewResolver
SC - Spring container
LVN - logical view name
AVN - Actual view name

Steps for Spring MVC

1. Copy n import from "Template Projects\spring_mvc_template" , to download spring MVC n hibernate dependencies, force update it.

2. Configure the entry point(common interceptor) - Front Controller , supplied by Spring web framework

o.s.w.s.DispatcherServlet -Can Intercept ANY request coming from ANY client

Configure it in web.xml

Important config

url-pattern /

servlet name spring

servlet-class o.s.w.s.DispatcherServlet

load-on-startup 1

3. **Job of D.S (DispatcherServlet)**

init() -> start the SC , in a dynamic web application.

Creates the instance of - XmlWebApplicationContext , represents SC within web application

Default xml config

file - **servletName-servlet.xml** (eg - spring-servlet.xml)

location - WEB-INF

consists of XML tags for launching the SC.

4. Create request handing controller (=handler) - prog supplied spring bean

Add class level annotation - @Controller

+

Add method level annotation - @RequestMapping (equivalent - service)

@GetMapping = RequestMapping + method=GET (doGet)

@PostMapping = RequestMapping + method=POST (doPost)

.....

5. **SC creates n populates map to store the entries**

Map i/f - HandlerMapping

Implementation class - RequestMappingHandlerMapping

Key - value of @RequestMapping

Value - F.Q controller cls Name + method name

It's a spring bean - singleton n eager

6. Using spring-servlet.xml , configure **ViewResolver spring bean.**

singleton n eager

Job - Translation from Logical view Name (LVN) -> Actual view Name(AVN)

, using the prefix n suffix properties.

AVN = prefix+LVN+suffix

7. All of above blocks (D.S , HandlerMapping-H.M , Handler , ViewResolver-V.R) are ready , right at the web app deployment time.

8. Create JSP based view layer

name - welcome.jsp
location - /WEB-INF/views

9. Test Spring MVC flow .

10. Render index page also using Spring MVC ,via D.S

Clnt sends URL - http://host:port/ctx_path/ , method=GET

-> intercepted by Front controller (D.S)

->D.S sends the key / to -> H.M -> sends back the value (HelloWorldController.renderIndexPage)

-> D.S invokes handler's method -> rets LVN to D.S

-> D.S sends it -> V.R ->rets AVN

-> D.S forwards the clnt to view layer

-> renders the view

11. Render results along with view layer.

How to share results between Handler n View layer

Ans - Using ModelAttributes

Handler sends the model attributes -> D.S -> adds them under request scope -> sends them to JSP

Access these using EL syntax.

12. How to add model attributes ?

12.1 Using o.s.w.s.ModelAndView class

Constructor

ModelAndView(String LVN , String modelAttrName,Object modelAttrValue)

OR

12.2Using o.s.ui.Model - i/f

Map of model attributes

Instead of handler method creating this Model map , simply ask SC to provide it. => IoC

HOW ? Add Model Map as the method argument , SC will create EMPTY model map n pass it to the method (D.I)

API for adding model attributes

public Model addAttribute(String attrName,Object attrValue)

13. Integration of Spring MVC with Hibernate

1. Un comment importing of hibernate-persistence.xml from Matster spring config file(i.e spring-servlet.xml)

2. To be continued....

